

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: CRISISLTLSUM | Context: Ecuador — Humanitarian Crisis Social Media Analysts

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output ontology has two parts: a binary timeline-membership label defined by three conjunctive criteria — relevant, informative, not repetitive [Q58] — and a four-axis summarization rubric (Coherence, Accuracy, Coverage, Overall quality, each 1–5) [Q106, Q151]. The criteria are abstract enough to be conceptually portable, but their operationalization is English- and US-context-specific: 'informative' is defined as facts vs. 'personal points of view' [Q58], an annotation norm reflecting US MTurk-worker intuitions; the Coherence rubric explicitly references 'English errors' and English-language readability [Q153]. The deployment's relevance criteria for humanitarian coordinators (road access status, informal community broadcasts, displacement indicators, IGEPN alert level changes, damnificado counts at parroquia level — see operational_context_notes) are structurally richer than the benchmark's binary fact-vs-opinion split. The elicitation rated OO as HIGH priority precisely because relevance for humanitarian coordinators differs structurally from US annotator norms. Score is 2 rather than 1 because the four-axis summary rubric is conceptually adaptable to Spanish summary evaluation, even though its English anchoring would need to be replaced.

ID	Evidence
Q58	"A tweet is labeled as part of the timeline only if it satisfies all the following three conditions: relevant: talks about the same event indicated in the seed tweet, informative: provides facts about the event but not only contains personal points of view, not repetitive: brings in new information." (p.5)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)
Q151	"During this task, you will be reading a timeline that consists of a sequence of tweets describing an event and eight different summaries for the event in the timeline. You will rate the quality of each of the eight summaries by four axes: coherence, accuracy, coverage, and overall quality." (p.21)
Q153	"For each summary, answer the question 'how coherent is the summary on its own?' on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)

Strengths

- The four-axis summary rubric (Coherence, Accuracy, Coverage, Overall) [Q106, Q151] is a reasonable, decomposable structure that could be re-anchored to Spanish humanitarian reporting norms; the explicit decomposition of relevance into three criteria [Q58] makes the operational definition transparent and auditable.

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Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: The binary relevance label and four-axis summary rubric are conceptually generic but operationally tuned to US English. Categories like 'informative' (facts vs. personal opinion) [Q58] do not capture humanitarian-coordinator distinctions such as 'logistical' vs. 'needs-assessment' vs. 'institutional-coordination' relevance [operational_context_notes].

OO-2: Missing categories for the deployment: road/bridge access status, informal community broadcast signals, displacement movement indicators, institutional coordination signals, casualty/damnificado counts at parroquia/cantón level, infrastructure damage to service delivery, official alert level changes (IGEPN color codes, SNGR declarations) — these are flagged in operational_context_notes as the actual relevance criteria for Ecuadorian humanitarian coordinators.

OO-3: The Coherence rubric explicitly references 'English errors' and English-language comprehensibility [Q153], embedding a non-portable language assumption. The 'fact vs. personal point of view' distinction [Q58] reflects US journalistic norms and may misclassify community broadcasts (often first-person but operationally critical) as non-informative.

OO-4: The elicitation indicates stakeholder-driven taxonomy redesign is necessary: an inter-operational team of Ecuadorian field coordinators, social analysts, and domain experts would need to define multi-class relevance categories aligned with humanitarian coordination workflows.

OO-5: Documented: structural-validity concern (binary relevance is too coarse for humanitarian operational salience); content-validity concern (key humanitarian categories missing); external-validity concern (English-anchored coherence does not generalize to Spanish summary evaluation).

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WEB-21	HumVI review notes humanitarian event detection datasets disproportionately English; structural relevance categories for humanitarian work are underrepresented (https://arxiv.org/html/2410.06370)
Q153	"For each summary, answer the question 'how coherent is the summary on its own?' on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21)
Q106	"Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)

Information Gaps

- No documented mapping from humanitarian-coordinator relevance categories (logistics, needs, displacement, coordination, casualties) to a multi-class extraction taxonomy exists.

Requires Expert Verification

- The full set of relevance categories operationally used by Cruz Roja Ecuatoriana / OCHA Ecuador in tweet-stream analysis, and whether they are encoded in any internal SOPs.
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Remediation

Gap 1: Binary fact-vs-opinion relevance and English-anchored coherence rubric do not capture humanitarian-coordinator relevance categories or Spanish-language summary quality.

Recommendation: Redesign relevance as a multi-class taxonomy (logistics/access, needs, displacement, coordination, alerts, casualties) elicited from Cruz Roja Ecuatoriana and OCHA Ecuador SOPs; replace 'English errors' coherence anchor with Spanish-language fluency anchored to OCHA/Cruz Roja reporting genre conventions.